

DATA MODELS

Data models define how the logical structure of a database is modeled. Data Models are fundamental entities to introduce abstraction in a DBMS. Data models define how data is connected to each other and how they are processed and stored inside the system.

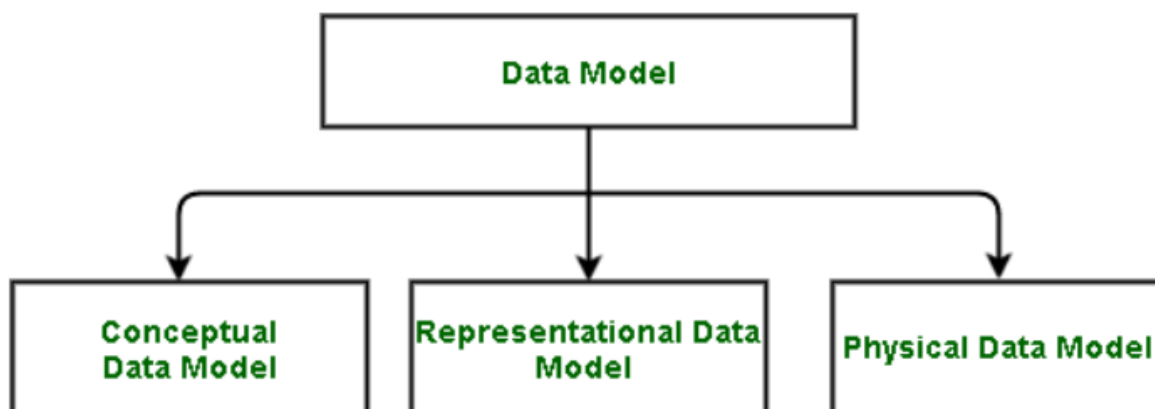
The very first data model could be flat data-models, where all the data used are to be kept in the same plane. Earlier data models were not so scientific, hence they were prone to introduce lots of duplication and update anomalies.

A Data Model in Database Management System (DBMS) is the concept of tools that are developed to summarize the description of the database. Data Models provide us with a transparent picture of data which helps us in creating an actual database. It shows us from the design of the data to its proper implementation of data.

Types of Relational Models

1. Conceptual Data Model
2. Representational Data Model
3. Physical Data Model

It is basically classified into 3 types: -



1. Conceptual Data Model

The conceptual data model describes the database at a very high level and is useful to understand the needs or requirements of the database. It is this model, that is used in the requirement-gathering process i.e. before the Database Designers start making a particular database. One such popular model is the entity/relationship model (ER model). The E/R model specializes in entities, relationships, and even attributes that are used by database designers.

Entity-Relationship Model(ER Model): It is a high-level data model which is used to define the data and the relationships between them. It is basically a conceptual design of any database which is easy to design the view of data.

Components of ER Model:

1. **Entity:** An entity is referred to as a real-world object. It can be a name, place, object, class, etc. These are represented by a rectangle in an ER Diagram.
2. **Attributes:** An attribute can be defined as the description of the entity. These are represented by Ellipse in an ER Diagram. It can be Age, Roll Number, or Marks for a Student.

3. **Relationship:** Relationships are used to define relations among different entities. Diamonds and Rhombus are used to show Relationships.

Characteristics of a conceptual data model

- Offers Organization-wide coverage of the business concepts.
- This type of Data Models are designed and developed for a business audience.
- The conceptual model is developed independently of hardware specifications like data storage capacity, location or software specifications like DBMS vendor and technology. The focus is to represent data as a user will see it in the “real world.”

Conceptual data models known as Domain models create a common vocabulary for all stakeholders by establishing basic concepts and scope

2. Representational Data Model

This type of data model is used to represent only the logical part of the database and does not represent the physical structure of the database. The representational data model allows us to focus primarily, on the design part of the database. A popular representational model is a Relational model. The relational Model consists of Relational Algebra and Relational Calculus. In the Relational Model, we basically use tables to represent our data and the relationships between them. It is a theoretical concept whose practical implementation is done in Physical Data Model.

The advantage of using a Representational data model is to provide a foundation to form the base for the Physical model.

3. Physical Data Model

The physical Data Model is used to practically implement Relational Data Model. Ultimately, all data in a database is stored physically on a secondary storage device such as discs and tapes. This is stored in the form of files, records, and certain other data structures. It has all the information on the format in which the files are present and the structure of the databases, the presence of external data structures, and their relation to each other. Here, we basically save tables in memory so they can be accessed efficiently. In order to come up with a good physical model, we have to work on the relational model in a better way. Structured Query Language (SQL) is used to practically implement Relational Algebra.

This Data Model describes **HOW** the system will be implemented using a specific DBMS system. This model is typically created by DBA and developers. The purpose is actual implementation of the database.

Characteristics of a physical data model:

- The physical data model describes data need for a single project or application though it maybe integrated with other physical data models based on project scope.
- Data Model contains relationships between tables that which addresses cardinality and nullability of the relationships.
- Developed for a specific version of a DBMS, location, data storage or technology to be used in the project.
- Columns should have exact datatypes, lengths assigned and default values.
- Primary and Foreign keys, views, indexes, access profiles, and authorizations, etc. are defined.

Some Other Data Models

1. Hierarchical Model

The hierarchical Model is one of the oldest models in the data model which was developed by IBM, in the 1950s. In a hierarchical model, data are viewed as a collection of tables, or we can say segments that form a hierarchical relation. In this, the data is organized into a tree-like structure where each record consists of one parent record and many children. Even if the segments are connected as a chain-like structure by logical associations, then the instant structure can be a fan structure with multiple branches. We call the illogical associations as directional associations.

2. Network Model

The Network Model was formalized by the Database Task group in the 1960s. This model is the generalization of the hierarchical model. This model can consist of multiple parent segments and these segments are grouped as levels but there exists a logical association between the segments belonging to any level. Mostly, there exists a many-to-many logical association between any of the two segments.

3. Object-Oriented Data Model

In the Object-Oriented Data Model, data and their relationships are contained in a single structure which is referred to as an object in this data model. In this, real-world problems are represented as objects with different attributes. All objects have multiple relationships between them. Basically, it is a combination of Object Oriented programming and a Relational Database Model.

4. Float Data Model

The float data model basically consists of a two-dimensional array of data models that do not contain any duplicate elements in the array. This data model has one drawback it cannot store a large amount of data that is the tables can not be of large size.

5. Context Data Model

The Context data model is simply a data model which consists of more than one data model. For example, the Context data model consists of ER Model, Object-Oriented Data Model, etc. This model allows users to do more than one thing which each individual data model can do.

6. Semi-Structured Data Model

Semi-Structured data models deal with the data in a flexible way. Some entities may have extra attributes and some entities may have some missing attributes. Basically, you can represent data here in a flexible way.

Advantages of Data Models

1. Data Models help us in representing data accurately.
2. It helps us in finding the missing data and also in minimizing Data Redundancy.
3. Data Model provides data security in a better way.
4. The data model should be detailed enough to be used for building the physical database.
5. The information in the data model can be used for defining the relationship between tables, primary and foreign keys, and stored procedures.

Disadvantages of Data Models

1. In the case of a vast database, sometimes it becomes difficult to understand the data model.
2. You must have the proper knowledge of [SQL](#) to use physical models.
3. Even smaller change made in structure require modification in the entire application.
4. There is no set data manipulation language in DBMS.
5. To develop Data model one should know physical data stored characteristics.

Introduction of ER Model

The Entity Relational Model is a model for identifying entities to be represented in the database and representation of how those entities are related. The ER data model specifies enterprise schema that represents the overall logical structure of a database graphically.

The Entity Relationship Diagram explains the relationship among the entities present in the database. ER models are used to model real-world objects like a person, a car, or a company and the relation between these real-world objects. In short, the ER Diagram is the structural format of the database.

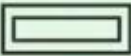
Why Use ER Diagrams In DBMS?

- ER diagrams are used to represent the E-R model in a database, which makes them easy to convert into relations (tables).
- ER diagrams provide the purpose of real-world modeling of objects which makes them intently useful.
- ER diagrams require no technical knowledge and no hardware support.
- These diagrams are very easy to understand and easy to create even for a naive user.
- It gives a standard solution for visualizing the data logically.

Symbols Used in ER Model

ER Model is used to model the logical view of the system from a data perspective which consists of these symbols:

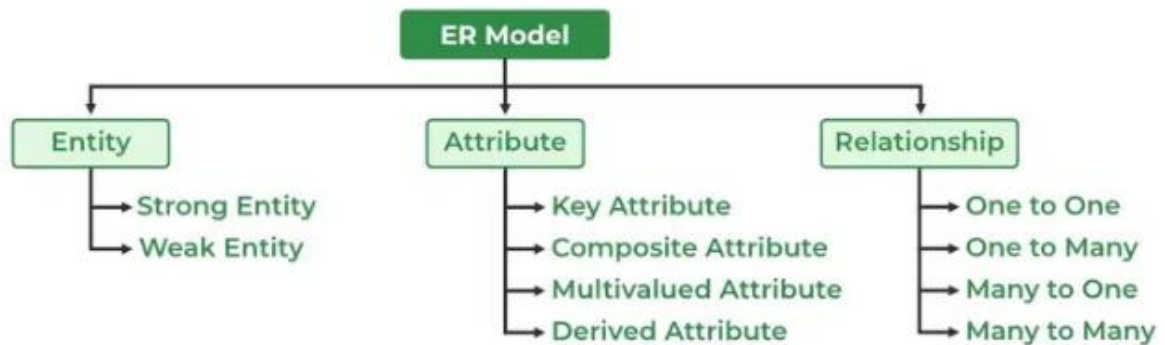
- **Rectangles:** Rectangles represent Entities in the ER Model.
- **Ellipses:** Ellipses represent Attributes in the ER Model.
- **Diamond:** Diamonds represent Relationships among Entities.
- **Lines:** Lines represent attributes to entities and entity sets with other relationship types.
- **Double Ellipse:** Double Ellipses represent [Multi-Valued Attributes](#).
- **Double Rectangle:** Double Rectangle represents a Weak Entity.

Figures	Symbols	Represents
Rectangle		Entities in ER Model
Ellipse		Attributes in ER Model
Diamond		Relationships among Entities
Line		Attributes to Entities and Entity Sets with Other Relationship Types
Double Ellipse		Multi-Valued Attributes
Double Rectangle		Weak Entity

Symbols used in ER Diagram

Components of ER Diagram

ER Model consists of Entities, Attributes, and Relationships among Entities in a Database System.



Components of ER Diagram

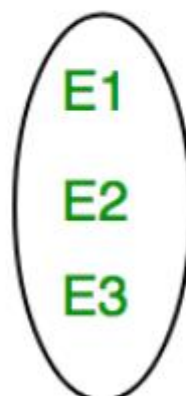
Entity

An Entity may be an object with a physical existence – a particular person, car, house, or employee – or it may be an object with a conceptual existence – a company, a job, or a university course.

Entity Set: An Entity is an object of Entity Type and a set of all entities is called an entity set. For Example, E1 is an entity having Entity Type Student and the set of all students is called Entity Set. In ER diagram, Entity Type is represented as:



Entity Type



Entity Set

Entity Set

1. Strong Entity

A Strong Entity is a type of entity that has a key Attribute. Strong Entity does not depend on other Entity in the Schema. It has a primary key, that helps in identifying it uniquely, and it is represented by a rectangle. These are called Strong Entity Types.

2. Weak Entity

An Entity type has a key attribute that uniquely identifies each entity in the entity set. But some entity type exists for which key attributes can't be defined. These are called Weak Entity types.

For Example, A company may store the information of dependents (Parents, Children, Spouse) of an Employee. But the dependents can't exist without the employee. So Dependent will be a **Weak Entity Type** and Employee will be Identifying Entity type for Dependent, which means it is **Strong Entity Type**.

A weak entity type is represented by a Double Rectangle. The participation of weak entity types is always total. The relationship between the weak entity type and its identifying strong entity type is called identifying relationship and it is represented by a double diamond.



Attributes

Attributes are the properties that define the entity type. For example, Roll_No, Name, DOB, Age, Address, and Mobile_No are the attributes that define entity type Student. In ER diagram, the attribute is represented by an oval.



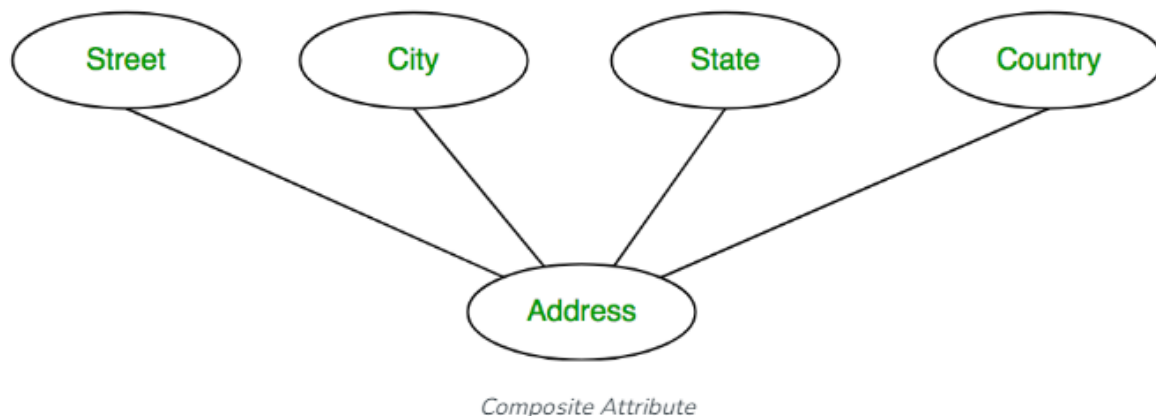
1. Key Attribute

The attribute which **uniquely identifies each entity** in the entity set is called the key attribute. For example, Roll_No will be unique for each student. In ER diagram, the key attribute is represented by an oval with underlying lines.



2. Composite Attribute

An attribute **composed of many other attributes** is called a composite attribute. For example, the Address attribute of the student Entity type consists of Street, City, State, and Country. In ER diagram, the composite attribute is represented by an oval comprising of ovals.



3. Multivalued Attribute

An attribute consisting of more than one value for a given entity. For example, Phone_No (can be more than one for a given student). In ER diagram, a multivalued attribute is represented by a double oval.

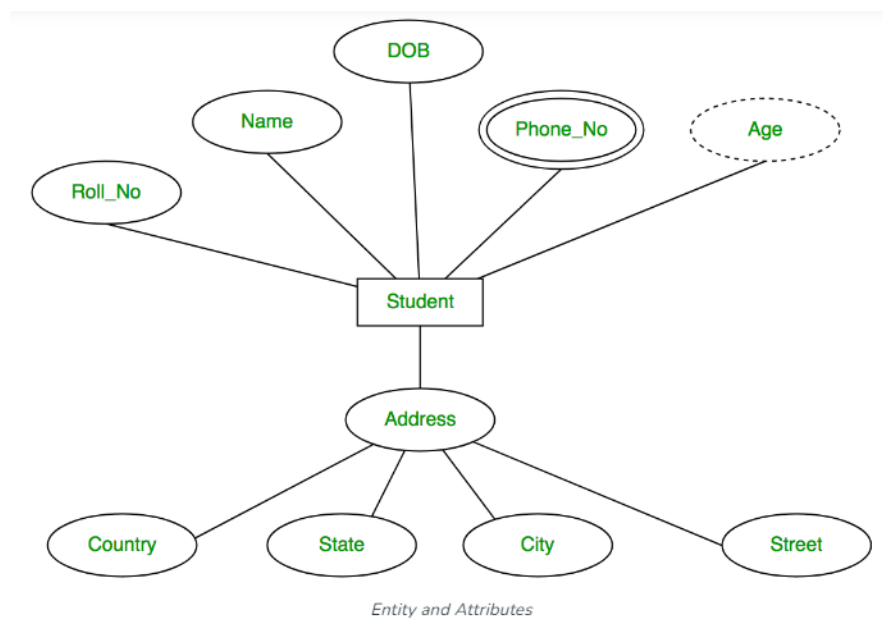


4. Derived Attribute

An attribute that can be derived from other attributes of the entity type is known as a derived attribute. e.g.; Age (can be derived from DOB). In ER diagram, the derived attribute is represented by a dashed oval.

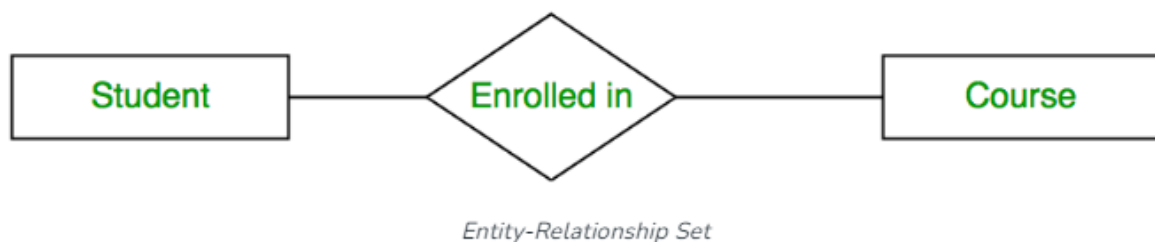


The Complete Entity Type Student with its Attributes can be represented as:

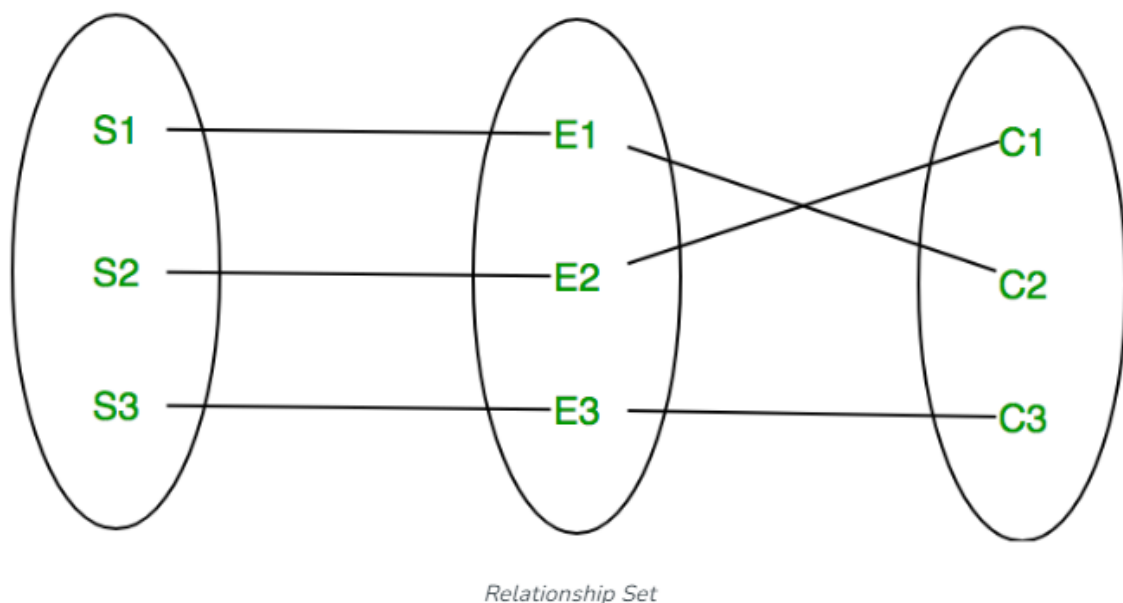


Relationship Type and Relationship Set

A Relationship Type represents the association between entity types. For example, ‘Enrolled in’ is a relationship type that exists between entity type Student and Course. In ER diagram, the relationship type is represented by a diamond and connecting the entities with lines.



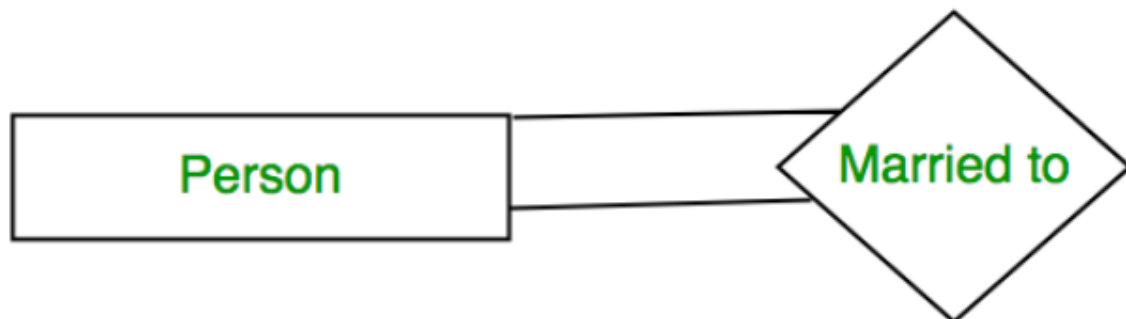
A set of relationships of the same type is known as a relationship set. The following relationship set depicts S1 as enrolled in C2, S2 as enrolled in C1, and S3 as registered in C3.



Degree of a Relationship Set

The number of different entity sets participating in a relationship set is called the degree of a relationship set.

1. Unary Relationship: When there is only ONE entity set participating in a relation, the relationship is called a unary relationship. For example, one person is married to only one person.



Unary Relationship

2. Binary Relationship: When there are TWO entities set participating in a relationship, the relationship is called a binary relationship. For example, a Student is enrolled in a Course.



Binary Relationship

3. Ternary Relationship: When there are n entities set participating in a relation, the relationship is called an n-ary relationship.

Cardinality

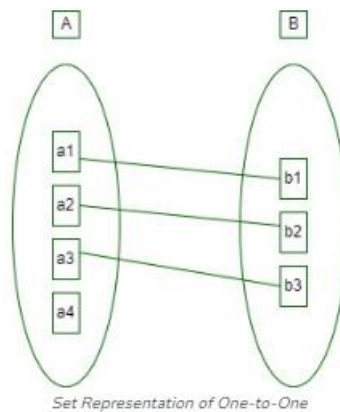
The number of times an entity of an entity set participates in a relationship set is known as cardinality. Cardinality can be of different types:

1. One-to-One: When each entity in each entity set can take part only once in the relationship, the cardinality is one-to-one. Let us assume that a male can marry one female and a female can marry one male. So the relationship will be one-to-one. the total number of tables that can be used in this is 2.

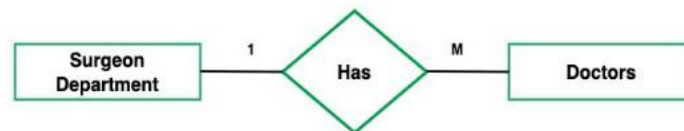


one to one cardinality

Using Sets, it can be represented as:

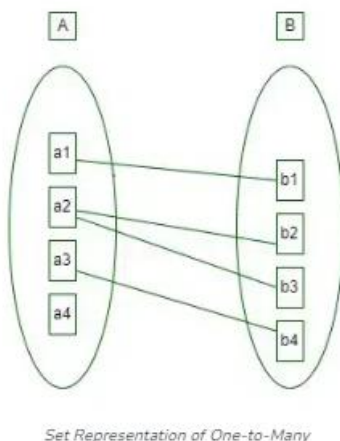


2. One-to-Many: In one-to-many mapping as well where each entity can be related to more than one entity and the total number of tables that can be used in this is 2. Let us assume that one surgeon department can accommodate many doctors. So the Cardinality will be 1 to M. It means one department has many Doctors.
total number of tables that can be used is 3.



one to many cardinality

Using sets, one-to-many cardinality can be represented as:

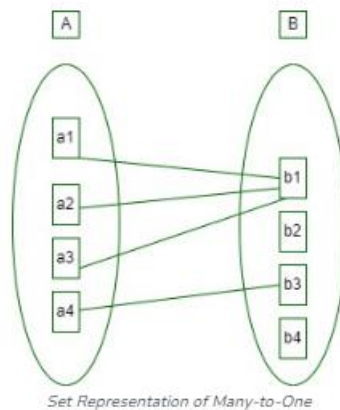


3. Many-to-One: When entities in one entity set can take part only once in the relationship set and entities in other entity sets can take part more than once in the relationship set, cardinality is many to one. Let us assume that a student can take only one course but one course can be taken by many students. So the cardinality will be n to 1. It means that for one course there can be n students but for one student, there will be only one course.
The total number of tables that can be used in this is 3.



many to one cardinality

Using Sets, it can be represented as:



Set Representation of Many-to-One

In this case, each student is taking only 1 course but 1 course has been taken by many students.

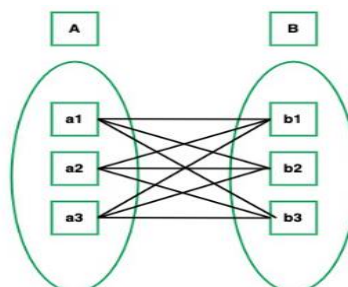
4. Many-to-Many: When entities in all entity sets can take part more than once in the relationship cardinality is many to many. Let us assume that a student can take more than one course and one course can be taken by many students. So the relationship will be many to many.

the total number of tables that can be used in this is 3.



many to many cardinality

Using Sets, it can be represented as:



Many-to-Many Set Representation

In this example, student S1 is enrolled in C1 and C3 and Course C3 is enrolled by S1, S3, and S4. So it is many-to-many relationships.

Difference between entity, entity set and entity type

Entity	Entity Type	Entity Set
A thing in the real world with independent existence	A category of a particular entity	Set of all entities of a particular entity type.
Any particular row (a record) in a relation(table) is known as an entity.	The name of a relation (table) in RDBMS is an entity type	All rows of a relation (table) in RDBMS is entity set

Difference between Strong and Weak Entity

Strong Entity	Weak Entity
Strong entity always has a primary key.	While a weak entity has a partial discriminator key.
Strong entity is not dependent on any other entity.	Weak entity depends on strong entity.
Strong entity is represented by a single rectangle.	Weak entity is represented by a double rectangle.
Two strong entity's relationship is represented by a single diamond.	While the relation between one strong and one weak entity is represented by a double diamond.
Strong entities have either total participation or not.	a weak entity always has total participation.

SOME EXAMPLES OF ER DIAGRAM

Construct an ER diagram for library management system.

Step 1: The first step is to identify the entity sets. These are things that have more than one instance of being existent. The entity sets for a library management system are as follows:

- Member
- Book
- Granter
- Section
- Publisher

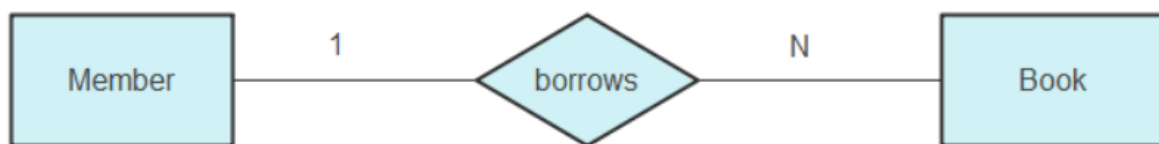
Step 2: The second step is to associate attributes to the entity sets. This is important because entity sets are recurring, such as several books, members, etc. The attributes are needed to differentiate between entities. The respective attributes are as follows:

- Member: Member ID, Name, Birthday, Address, Age
- Book: Author, ISBN, Title, Author, Price
- Granter: Name, NIC, Number, Address, Post
- Section: SID, Name, Phone
- Publisher: PID, Name, Address, Phone

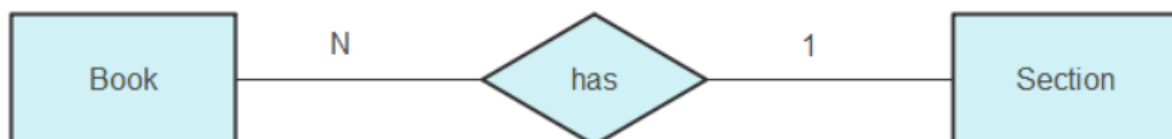
Step 3: The third step is to find the key attributes of each entity. The key attribute is unique to every single entity component. The key attributes are as follows:

- Member – Member ID
- Book – ISBN
- Granter – NIC
- Section – SID
- Publisher – PID

Step 4: The fourth step is to identify the relationship between the different entities present. This is necessary to analyze the database clearly and make a better library management system.



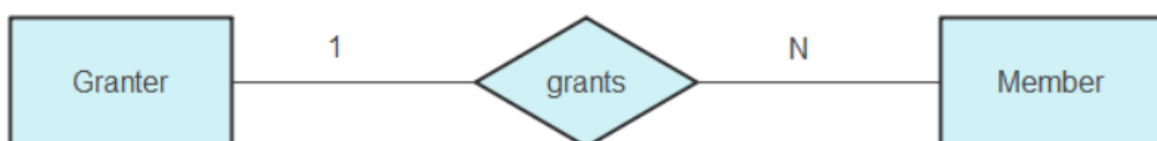
A single member can borrow multiple books if they want. This kind of relationship is known as a one-to-many relationship.



In this relation, a single section can have multiple books present over there. Making this a one-to-many relationship as well.

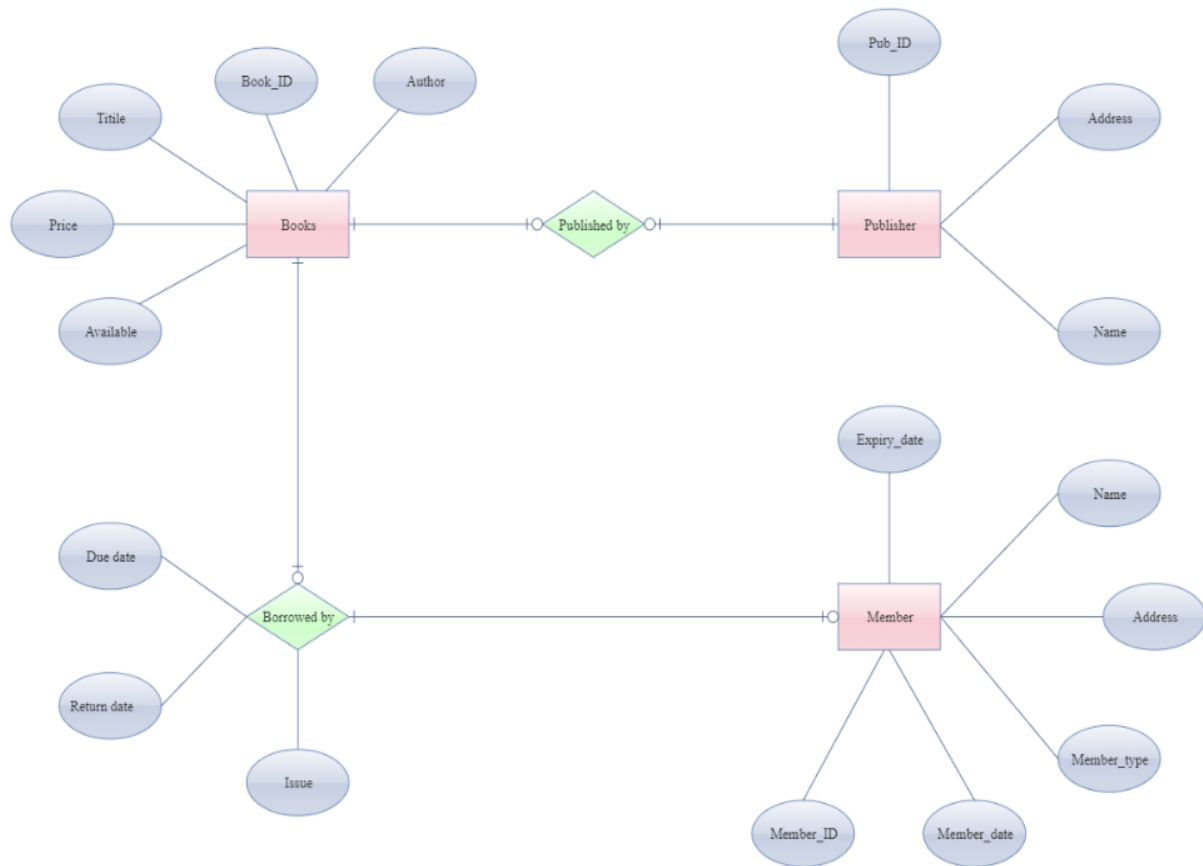


A single publisher can publish several books and supply them to the library. This is another one-to-many relationship.



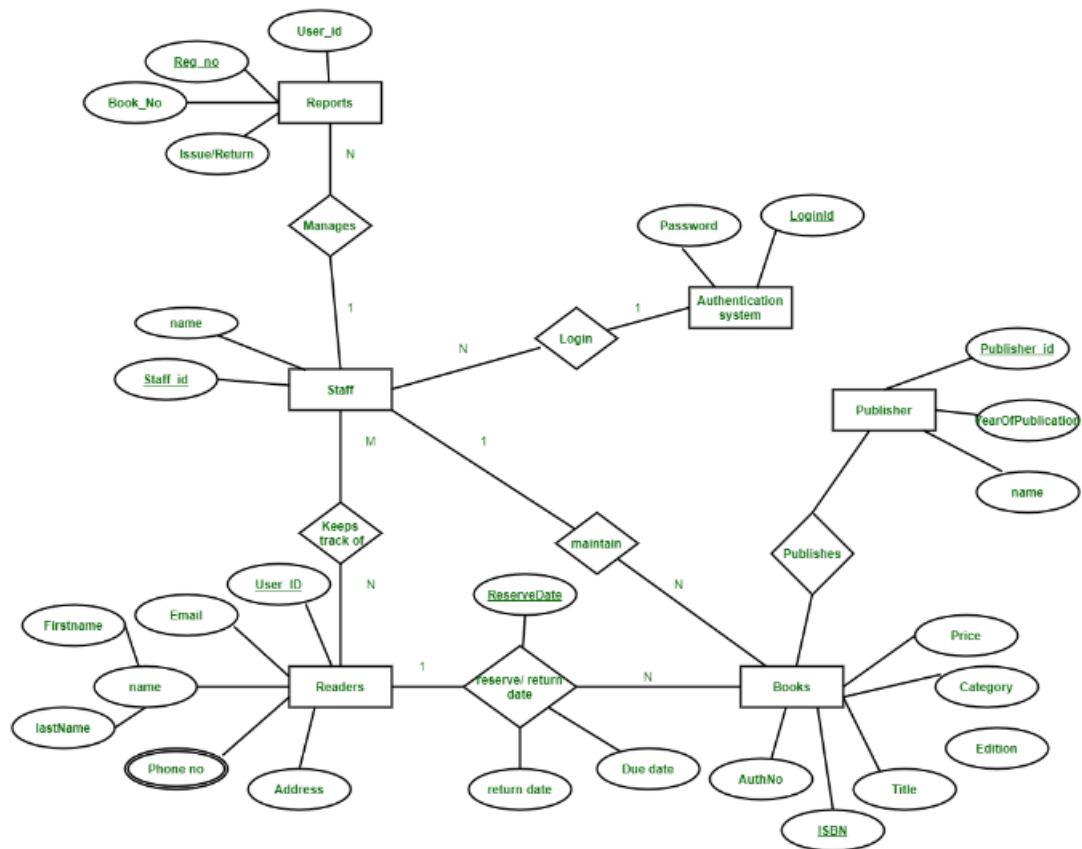
The granter grants memberships to the student. There is only one granter and multiple members; thus, this is a one-to-many relationship.

Step 5: The fifth step is to compile all these steps. Join the entities that go together and form an ER diagram.



OR

Note: Draw ER Diagram by step by step process



ER DIAGRAM FOR UNIVERSITY MANAGEMENT SYSTEM

